

The sharp constant of the dyadic triangular Hilbert form is at least 3

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Abstract

Let $\Lambda(F_0, F_1, F_2) = \sum_{k \in \mathbb{Z}} \sum_{I \in A_k} |L_I(F_0, F_1, F_2)|$ be the dyadic triangular Hilbert form with absolute local contributions, in the normalisation of the recent preprint *An L^3 bound for the dyadic triangular Hilbert form* (OpenAI, 2026), which proves $\Lambda \leq 40 \prod_v \|F_v\|_{L^3(\mathbb{R}_+^2)}$ for real inputs. Let C^* be the best constant. We prove $C^* \geq 3$. The proof is elementary: for inputs constant on dyadic squares, Λ is a finite sum of integer-valued traces, and an exact *product rule* describes how it transforms under Kronecker (tensor) products of inputs. Tensor powers of one explicit 4×4 sign pattern g with $(H, T) = (\frac{3}{2}, \frac{1}{2})$ then have ratio exactly $3 - 2^{1-r}$, so the supremum is at least 3. All values are exact rationals, computed by two independent implementations and, for the first instance beyond $5/2$, by a third. We also prove that the sharp constant at resolution 1 is exactly 1 and that each dyadic level contributes at most $\prod_v \|F_v\|_3$. We conjecture $C^* = 3$, and state the conjecture in a sharper, tensor-closed form, $H(F) + 3|T(F)| \leq 3 \prod_v \|F_v\|_3$, which is saturated on the whole tensor family. Exhaustive search at resolution 1 and exact hill-climbs at resolutions 2–4 found no violation. The gap $3 \leq C^* \leq 40$ remains. Nothing here is machine-checked in a proof assistant.

1 Introduction

The triangular Hilbert transform couples three functions of two variables, each pair sharing one variable, and its $L^3 \times L^3 \times L^3$ estimate is a well-known open problem of multilinear harmonic analysis [3, 2]. A dyadic model, in which the intervals are constrained by binary addition without carries, was formulated alongside it. The preprint [1] (one of the manuscripts released in the `openai/math` repository, produced by an internal OpenAI model) proves the uniform estimate for this dyadic model with unrestricted real inputs and the explicit constant 40. That manuscript is not formalised in Lean and we have not verified its proof; we use only its *definition* of the form, which we restate in full.

The question we address is the size of the best constant. Our contribution is a lower bound.

Theorem 1.1. *Let C^* be the best constant in $\Lambda(F_0, F_1, F_2) \leq C^* \prod_v \|F_v\|_{L^3(\mathbb{R}_+^2)}$ over real, bounded, compactly supported F_v . Then $C^* \geq 3$.*

The mechanism is a product rule (Theorem 3.1): on inputs constant on dyadic squares, the form of a Kronecker product $f \otimes g$ is an affine function of the form of f with slope $|T(g)|$, the top term of g . A seed with $H(g)/(1 - |T(g)|) = 3$ therefore generates a sequence with ratios $3 - 2^{1-r}$. The value $5/2$ found earlier by numerical ascent (Section 8) is the second member of that sequence.

2 Definitions and the finite reduction

We write $\mathbb{R}_+ = [0, \infty)$. For $k \in \mathbb{Z}$ let $D_k = \{[2^k n, 2^k(n+1)) : n \in \mathbb{Z}_{\geq 0}\}$, and for $I \in D_k$ with index n_I let $h_I = \mathbf{1}_{I_{\text{left}}} - \mathbf{1}_{I_{\text{right}}}$. Let $\oplus$ be bitwise XOR and

$$A_k = \{(I_0, I_1, I_2) \in D_k^3 : n_{I_0} \oplus n_{I_1} \oplus n_{I_2} = 0\}.$$

For real measurable F_0, F_1, F_2 on $\mathbb{R}_+^2$ and $I \in A_k$,

$$L_I(F_0, F_1, F_2) = 2^{-k} \int_{I_0 \times I_1 \times I_2} F_0(x_0, x_1) F_1(x_1, x_2) F_2(x_2, x_0) h_{I_0}(x_0) h_{I_1}(x_1) h_{I_2}(x_2) dx_0 dx_1 dx_2, \quad (1)$$

and $\Lambda(F) = \sum_{k \in \mathbb{Z}} \sum_{I \in A_k} |L_I(F)|$. Theorem 1.1 of [1] states $\Lambda(F) \leq 40 \prod_v \|F_v\|_3$ for real bounded compactly supported inputs, and extends to L^3 by density. We write $R(F) = \Lambda(F) / \prod_v \|F_v\|_3$ and $C^* = \sup R$.

Step inputs. Fix $N \geq 1$, $n = 2^N$, and let each F_v be constant on the squares of side 2^{-N} inside $[0, 1)^2$ and zero elsewhere; identify F_v with the $n \times n$ matrix $F_v[i, j]$ of its values. Write $T(F) = \int F_0 F_1 F_2$ over the unit cube, so that $T = n^{-3} \sum_{i,j,k} F_0[i, j] F_1[j, k] F_2[k, i] = n^{-3} \text{tr}(F_0 F_1 F_2)$.

Proposition 2.1 (finite reduction). *For step inputs at resolution N ,*

$$\Lambda(F) = |T(F)| + H(F), \quad H(F) = \sum_{l=0}^{N-1} \sum_{I \in A_{-l}} |L_I(F)|,$$

and every L_I is $2^l n^{-3}$ times an integer combination of products of entries. Dyadic dilations leave R unchanged, so restricting the support to $[0, 1)^2$ loses nothing.

Proof. At scale $2^k \geq 2$ only the triple of intervals containing $[0, 1)$ meets the support, all three Haar functions equal $+1$ there, and $L = 2^{-k} T$; these scales sum to $|T| \sum_{k \geq 1} 2^{-k} = |T|$. At scale $2^k \leq 2^{-N}$ each h_I integrates to zero on every cell, so $L_I = 0$. At level l , i.e. $k = -l$ with $0 \leq l \leq N-1$, there are $m = 2^l$ intervals of $b = n/m$ cells each; writing $s \in \{\pm 1\}^b$ for the Haar signs and F_v^{pq} for the $b \times b$ block of F_v on the interval pair (p, q) , the triple $(p, q, p \oplus q)$ contributes

$$L_I = 2^l n^{-3} \text{tr}(S F_0^{pq} S F_1^{q, p \oplus q} S F_2^{p \oplus q, p}), \quad S = \text{diag}(s). \quad (2)$$

Dilation invariance follows from the scaling of (1). $\square$

For sign inputs ($F_v[i, j] \in \{\pm 1\}$) every $\|F_v\|_3 = 1$, so $R = |T| + H$ is an exact rational. All computations below use integer arithmetic on (2); nothing is floating point.

3 The product rule

For $f = (f_0, f_1, f_2)$ at resolution N_1 and g at resolution N_2 , let $f \otimes g$ be the input at resolution $N_1 + N_2$ with $(f_v \otimes g_v)[i, j] = f_v[\lfloor i/2^{N_2} \rfloor, \lfloor j/2^{N_2} \rfloor] g_v[i \bmod 2^{N_2}, j \bmod 2^{N_2}]$ (the Kronecker product: f on the coarse cells, a copy of g inside each). Put

$$A'(f) = 2^{-2N_1} \sum_{p \oplus q \oplus r = 0} |f_0[p, q] f_1[q, r] f_2[r, p]|,$$

the triangular form of $|f|$ over the admissible coarse triples; $A'(f) = 1$ for sign inputs.

Theorem 3.1 (product rule). *For real step inputs f, g ,*

$$\Lambda(f \otimes g) = |T(f)| |T(g)| + |T(g)| H(f) + A'(f) H(g), \quad \|f_v \otimes g_v\|_3 = \|f_v\|_3 \|g_v\|_3, \quad T(f \otimes g) = T(f)T(g).$$

In particular, for sign inputs,

$$R(f \otimes g) = H(g) + |T(g)| R(f). \quad (3)$$

Proof. Let $F = f \otimes g$, $n_1 = 2^{N_1}$, $n_2 = 2^{N_2}$.

Coarse levels $l < N_1$. Each Haar function is constant on every coarse cell of side 2^{-N_1} , so the integral in (1) splits into a sum over coarse cell triples (a_0, a_1, a_2) of $\pm f_0[a_0, a_1] f_1[a_1, a_2] f_2[a_2, a_0]$ times the integral of $g_0 g_1 g_2$ over the cell cube, which is $n_1^{-3} T(g)$ after rescaling the cell to the unit cube. Hence $L_I(F) = T(g) L_I(f)$ for every coarse triple, and the same computation with all signs +1 gives $T(F) = T(f)T(g)$.

Fine levels $l \geq N_1$. Write $I_v = 2^{-N_1}(a_v + J_v)$ with J_v dyadic of level $l - N_1$. The index of I_v is $a_v 2^{l-N_1} + j_v$, so the XOR condition holds iff $a_0 \oplus a_1 \oplus a_2 = 0$ and $j_0 \oplus j_1 \oplus j_2 = 0$. On the cube of I the inputs equal $f_0[a_0, a_1] f_1[a_1, a_2] f_2[a_2, a_0]$ times the fine copy of g , the prefactor splits as $2^l = 2^{N_1} 2^{l-N_1}$ and the measure contributes n_1^{-3} , so $L_I(F) = f_0[a_0, a_1] f_1[a_1, a_2] f_2[a_2, a_0] \cdot n_1^{-2} L_J(g)$. Summing $|L_I(F)|$ over the admissible (a_0, a_1) and over J gives $A'(f) \sum_J |L_J(g)|$ at level $l - N_1$; over all fine levels this is $A'(f)H(g)$.

The norm identity is $\|f_v \otimes g_v\|_3^3 = n_1^{-2} n_2^{-2} \sum |f_v|^3 \sum |g_v|^3$. Adding the two groups of levels to $|T(F)|$ gives the formula; for sign inputs all norms and $A'(f)$ equal 1. $\square$

Lemma 3.2 (closure). *For sign inputs with $|T| < 1$ put $\Phi(g) = H(g)/(1 - |T(g)|)$. If $\Phi(f) \leq M$ and $\Phi(g) \leq M$ then $\Phi(f \otimes g) \leq M$; and $R(g^{\otimes r}) \rightarrow \Phi(g)$ monotonically from below as $r \rightarrow \infty$.*

Proof. By (3) applied twice, $H(f \otimes g) = H(g) + |T(g)|H(f)$ and $|T(f \otimes g)| = |T(f)||T(g)|$. Then $H(f \otimes g) \leq M(1 - |T(g)|) + |T(g)|M(1 - |T(f)|) = M(1 - |T(f)||T(g)|)$. For the second claim, $R_r = H + |T|R_{r-1}$ with $R_1 = H + |T|$ is an affine contraction with fixed point Φ , and $R_1 \leq \Phi$ iff $H + |T| \leq H/(1 - |T|)$, which holds. $\square$

Thus the tensor construction can never exceed the best Φ of a “prime” seed, and the lower bound it yields is exactly $\sup_g \Phi(g)$.

4 The lower bound

Theorem 4.1. *Let $g = (g_0, g_1, g_2)$ be the sign input at resolution 2*

$$g_0 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & -1 \end{pmatrix}, \quad g_1 = \begin{pmatrix} -1 & 1 & 1 & -1 \\ 1 & -1 & 1 & -1 \\ -1 & -1 & 1 & 1 \\ -1 & -1 & -1 & -1 \end{pmatrix}, \quad g_2 = \begin{pmatrix} -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \\ 1 & -1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{pmatrix}.$$

Then $H(g) = \frac{3}{2}$, $T(g) = \frac{1}{2}$, $R(g) = 2$, and $R(g^{\otimes r}) = 3 - 2^{1-r}$ for every $r \geq 1$. Hence $C^ \geq 3$.*

Proof. The values of H and T are finite integer computations from (2) (Section 7). By (3), $R_r = \frac{3}{2} + \frac{1}{2}R_{r-1}$ with $R_1 = 2$, whose solution is $3 - 2^{1-r}$. Each $g^{\otimes r}$ is a bounded compactly supported sign input, so $C^* \geq \sup_r R(g^{\otimes r}) = 3$. $\square$

Table 1 lists the exact values up to $r = 5$ (resolution $N = 10$), each computed directly from (2) rather than from the recursion, together with the closed form.

r	N	$R(g^{\otimes r})$	H	T	$3 - 2^{1-r}$	match
1	2	2	$\frac{3}{2}$	$\frac{1}{2}$	2	yes
2	4	$\frac{5}{2}$	$\frac{9}{4}$	$\frac{1}{4}$	$\frac{5}{2}$	yes
3	6	$\frac{11}{4}$	$\frac{21}{8}$	$\frac{1}{8}$	$\frac{11}{4}$	yes
4	8	$\frac{23}{8}$	$\frac{45}{16}$	$\frac{1}{16}$	$\frac{23}{8}$	yes
5	10	$\frac{47}{16}$	$\frac{93}{32}$	$\frac{1}{32}$	$\frac{47}{16}$	yes

Table 1: Exact ratios of the tensor powers of the seed g (direct evaluation, integer arithmetic).

Remark 4.2. The seed g is the sign pattern of the resolution-2 witness found by numerical ascent (Section 8); it attains the exhaustive resolution-2 maximum $R = 2$ among sign inputs (Section 7). Its tensor square $g \otimes g$ has ratio $5/2$ and explains the value first reached by floating-point ascent at resolution 4.

5 Upper bounds

Write C_N for the best constant over step inputs at resolution N (real entries). Clearly $C_1 \leq C_2 \leq \dots$ and $C^* = \sup_N C_N$ by density.

Proposition 5.1. *For real step inputs, $|T(F)| + \sum_{I \in A_0} |L_I(F)| \leq \prod_v \|F_v\|_3$ and, for each level l , $\sum_{I \in A_{-l}} |L_I(F)| \leq \prod_v \|F_v\|_3$. Consequently $C_N \leq N + 1$, and $C_1 = 1$.*

Proof. We use the triangular Hölder inequality: for any abelian group G and functions u_v on G^2 , $\sum_{x \oplus y \oplus z = 0} |u_0(x, y) u_1(y, z) u_2(z, x)| \leq \prod_v \|u_v\|_{\ell^3(G^2)}$, because the three maps $(x, y) \mapsto (x, y), (y, z), (z, x)$ from the admissible set to G^2 are bijections, so the generalised Hölder inequality with exponents $(3, 3, 3)$ applies and each factor is the full ℓ^3 norm. (The same holds for the odd class $x \oplus y \oplus z = 1$ and, with Lebesgue measure, for integrals over half-cubes.)

Level l . Each $|L_I| \leq 2^l \int_{I_0 \times I_1 \times I_2} |F_0 F_1 F_2|$. On that cube Hölder with exponents $(3, 3, 3)$ gives $\int |F_0 F_1 F_2| \leq \prod_v (2^{-l} \int_{Q_v} |F_v|^3)^{1/3} = 2^{-l} \prod_v \|F_v \mathbf{1}_{Q_v}\|_3$, where $Q_0 = I_0 \times I_1$, $Q_1 = I_1 \times I_2$, $Q_2 = I_2 \times I_0$ (the third variable ranges over an interval of length 2^{-l}). Summing over $I \in A_{-l}$ and applying the triangular Hölder inequality on the group $(\mathbb{Z}/2)^l$ of interval indices under XOR, with $u_v(p, q) = \|F_v \mathbf{1}_{I_p \times I_q}\|_3$, gives $\sum_I |L_I| \leq \prod_v (\sum_{p, q} \|F_v \mathbf{1}_{I_p \times I_q}\|_3^3)^{1/3} = \prod_v \|F_v\|_3$.

Top terms. $T \pm L_{(0,0,0)} = \int F_0 F_1 F_2 (1 \pm h(x_0)h(x_1)h(x_2))$, and $1 + hhh$ equals 2 on the four half-cubes of even parity and 0 otherwise (and $1 - hhh$ the reverse). So $|T| + |L_{(0,0,0)}| = \max(|T + L|, |T - L|) \leq 2 \max_{\text{parity}} \int_{\text{class}} |F_0 F_1 F_2|$, and on each half-cube Hölder gives a factor $\frac{1}{2} \prod_v \|F_v \mathbf{1}_{Q_v}\|_3$; the four half-cubes of one parity class map bijectively onto the four quarter-squares of each F_v , so the triangular Hölder inequality on $\mathbb{Z}/2$ bounds the sum by $\frac{1}{2} \prod_v \|F_v\|_3$. Hence $|T| + |L_{(0,0,0)}| \leq \prod_v \|F_v\|_3$.

Adding the level bounds gives $C_N \leq N + 1$; at $N = 1$ the only level is $l = 0$, so $C_1 \leq 1$, and $F \equiv 1$ attains 1. $\square$

The bound $C_N \leq N + 1$ is attained when all Haar signs are replaced by $+1$ (the Haar-free form), which is how the numerical instruments are checked for power. The uniform bound 40 of [1] exploits cancellation across levels through a matrix energy $\text{tr}((R^* R)^{3/2})$; we have not improved it.

6 A sharp conjecture

Conjecture 6.1. *For all real step inputs (equivalently all real inputs), $H(F) + 3|T(F)| \leq 3 \prod_v \|F_v\|_3$.*

Since $R = |T| + H \leq H + 3|T|$, Conjecture 6.1 implies $C^* \leq 3$, hence $C^* = 3$ by Theorem 4.1. The functional $\Psi(F) = H(F) + 3|T(F)|$ is the natural object for three reasons. First, for sign inputs $\Psi \leq 3$ is equivalent to $\Phi \leq 3$, so it is exactly the statement that no seed beats the one found. Second, equality propagates through tensor products: if $\Psi(f) = \Psi(g) = 3$ then, by Theorem 3.1, $\Psi(f \otimes g) = H(g) + |T(g)|(H(f) + 3|T(f)|) = H(g) + 3|T(g)| = 3$. The constant input ($H = 0, T = 1$) and the seed g ($H = \frac{3}{2}, T = \frac{1}{2}$) both satisfy $\Psi = 3$, so the entire family $g^{\otimes r}$, and every tensor product of copies of g and $\mathbf{1}$, lies on the equality set. Third, at resolution 1 it is a theorem: $\Psi \leq |L| + 3|T| \leq 1 + 2|T| \leq 3$ by Proposition 5.1. A proof for general N cannot proceed cell by cell, because the fine levels only see the admissible (even) coarse triples while T sees all eight, and each square of F_v is shared by one even and one odd triple; this coupling is where the content of the conjecture lies.

7 Computations

All results are in the repository `xaviercallens/AutoevolveAI` at commit `b4fa5cdee3a5`, directory `results/openai_math/hypotheses/tensor/`, produced by `scripts/openai_math/hypotheses/tensor/h5_tensor_rigor.py`, sha256 prefix `04b401f06bbf7405`; `phi_search.py`, `281e6038f6e4b1f8`), and archived at Zenodo (DOI 10.5281/zenodo.23232389).

Independent implementations. The evaluator in `h5_tensor_rigor.py` was written from (1)–(2) and agrees with the earlier evaluator `h5_exact.py` on random sign inputs at resolutions 1–4 (test suite). The value $\frac{23}{8}$ of $w_4 \otimes w_4$ (Section 8) was obtained by both, and by a third implementation written by an adversarial verifier agent from the definition, with a different contraction order ($\frac{23}{8}$). Controls: the constant input gives exactly 1; the Haar-free form gives exactly $N + 1$; the product rule was checked exactly on 8 random sign pairs and, in its general form with $A'(f)$, on 6 random rational pairs (all pass: yes).

Exhaustive resolution 1. Over all 4096 sign inputs, $\max R = 1$ and $\max \Phi = 1$.

Resolution 2. The exhaustive maximum of R over the 2^{48} sign inputs is 2. The computation (night lane of 2026-10-07) fixes (F_0, F_1) , reduces by the symmetries $F_v \rightarrow -F_v$, and uses that for fixed coefficient vectors c_t of the six terms, $\max_{F_2 \in \{\pm 1\}^{16}} \sum_t |\langle c_t, F_2 \rangle| = \max_{s \in \{\pm 1\}^6} \|\sum_t s_t c_t\|_1$; each chunk’s argmax was rechecked by the exact evaluator. The seed g attains this maximum.

Searches for a seed beating 3. Exact hill-climbs on Φ (single and pair sign flips, random and perturbed tensor starts) and on R found no value of Φ above 3 (best Φ : 3; best R : $\frac{5}{2}$; any above 3: none). Table 2 lists the runs. A hill-climb proves nothing; it is reported as the evidence it is.

8 How the bound was found

The constant was first studied by preregistered numerical experiments in a research pipeline run by language-model agents. A Hölder-ascent search at resolutions 2–7 reached exactly 2 at $N \leq 3$ and

objective	N	restarts	starts	best (exact)	best
R	3	200	random	2	2.0000
R	4	60	tensor + random	$\frac{5}{2}$	2.5000
Φ	2	400	random	3	3.0000
Φ	3	200	random	$\frac{103}{39}$	2.6410
Φ	3	120	tensor + random	$\frac{53}{21}$	2.5238
Φ	4	60	tensor + random	3	3.0000

Table 2: Exact sign-flip hill-climbs (`phi_search.py`). None exceeds 3.

$5/2$ at $N \geq 4$. The hypothesis $C^* = 2$ was preregistered and refuted by its own run; the amended hypothesis $C^* = 5/2$ survived fresh-seed runs at $N = 6, 7$ and was then refuted at $N = 8$ by the tensor square of the $N = 4$ witness w_4 , with exact ratio $\frac{23}{8}$. The product rule was found while explaining that number. Local ascent never found the tensor extremisers on its own: every ascent run at resolutions 4–8 converged to $5/2$ to within 10^{-10} and never left it, which is consistent with $5/2$ being a local maximum of R among the inputs reachable by small moves.

9 Limitations and AI use

The proofs are short. Theorem 3.1 was read and its identity re-derived numerically by an independent model verifier; Proposition 2.1, Lemma 3.2 and Proposition 5.1 were written for this manuscript and have been checked only by the agent that wrote them. None is machine-checked in a proof assistant. The exhaustive resolution-2 maximum is a computer-assisted claim from a preregistered run; the resolution-1 maximum was recomputed here. The upper bound 40 is taken from [1] and is not verified here; Theorem 1.1 does not depend on it. Whether the true constant is 3 is open.

The computations, the search for the seed, the product rule and this manuscript were produced by Claude agents (Anthropic) running in Claude Code under a workflow that the author designed and launched: preregistration committed before each confirmatory run, positive and negative controls, exact arithmetic, and an adversarial model verifier for every lane. The author chose the problem and the rules. No person has independently refereed this manuscript before this preprint.

References

- [1] OpenAI, *An L^3 bound for the dyadic triangular Hilbert form*, preprint, 5 October 2026, in github.com/openai/math (commit `adc7f124`), [preprints/An-L3-bound-for-the-dyadic-triangular-Hilbert-form-October-5-2026/](https://arxiv.org/abs/2610.00000).
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